Operational Research Tutorial

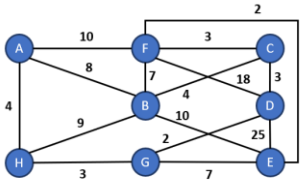
Networks – Minimum Spanning Tree and Shortest Paths

Algorithms

Minimum Spanning Tree

Prim's

1. Initialise distances to vertices as ∞ and predecessors as –
2. Arbitrarily select a vertex and label as selected
3. Update distances and predecessors to adjacent unselected vertices
4. Select the reachable vertex with minimum distance
5. Repeat 3 and 4 until all vertices have been selected

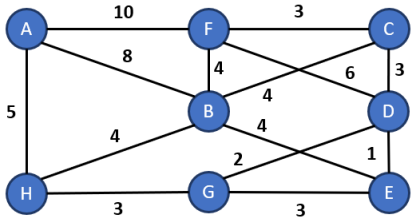


v	Selected	d_v	π_v
A			
B			
C			
D			
E			
F			
G			
H			

Kruskal's

1. Order the edges by weightings, lowest to highest
2. Select the lowest-weight edge not yet considered; if it does not form a cycle, label it as selected
3. Repeat step 2 until all vertices have been selected

Kruskal's Algorithm



Select first $|V| - 1$ edges that do not form a cycle

Edge e	w_e	Selected
(D,E)	1	
(D,G)	2	
(E,G)	3	
(C,D)	3	
(G,H)	3	
(C,F)	3	
(B,C)	4	
(B,E)	4	
(B,F)	4	
(B,H)	4	
(A,H)	5	
(D,F)	6	
(A,B)	8	
(A,F)	10	

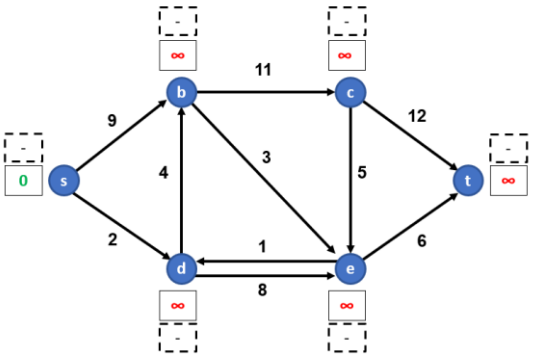
Shortest Path

Dijkstra's

1. Initialise the network, source vertex has a distance of 0 and all other vertices a distance of ∞, label all vertices as **temporary**
2. Label the vertex with minimum distance as **permanent**
3. Update the distances and predecessors to reachable **temporary** vertices from the current permanent vertex
4. Repeat steps 2 and 3 until all vertices are marked **permanent**

Sensitivity: Internal

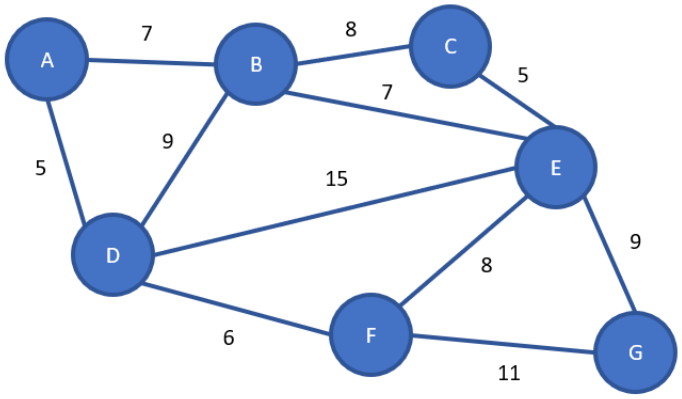
v	Label	d_v	π_v
s	T	0	-
b	T		
c	T		
d	T		
e	T		
t	T		



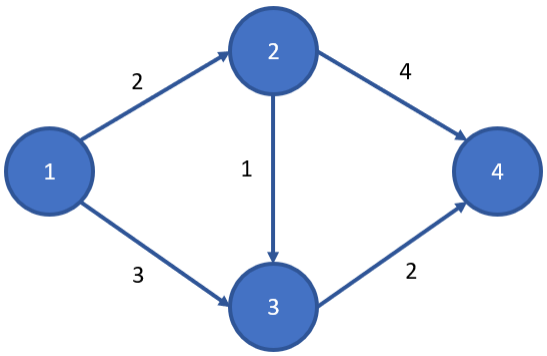
Sensitivity: Internal

Exercises

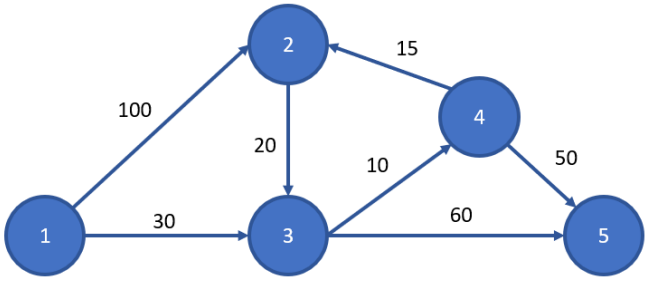
1. For the following network compute the minimum spanning tree using Prim's and Kruskal's Algorithms.



2. For the following networks:
   a. Compute the shortest path using Dijkstra's algorithm from node 1 to node 4, state the path and its length.



   b. Compute the shortest path using Dijkstra's algorithm from node 1 to node 2, state the path and its length.



Sensitivity: Internal